

Predicting Earth-Like Similarity in Exoplanets using Machine Learning Experimental Results

By: Samia Mahmood

Experiment #1: Baseline Ranking Comparison Study

The purpose of the Experiment #1: Baseline Ranking Comparison Study is to show that our proposed method weighted Earth-like similarity ranking provides more meaningful results than simpler ranking approaches. These simpler baseline K-Nearest Neighbor (KNN) models will be unweighted where all feature weights = 1. This will allow us to determine if the same strong Earth-like candidates remain near the top even when simpler methods are used. If the top-ranked candidates remain stable, that supports the robustness of the method. If some rankings change, we can analyze which features or weighting choices caused the difference.

Therefore, our weighted proposed method KNN models #1 and #2 will be compared to unweighted baseline KNN models #3 and #4 using the Manhattan and Euclidean distance metrics respectively. Then the same weighted proposed method KNN models #1 and #2 will be compared to unweighted baseline KNN models #5 and #6 which is filtered for exoplanets whose habitable zone flag = 1. This will also use the Manhattan and Euclidean distance metrics respectively. The comparison of the KNN models #5 and #6 will allow us to isolate the effect of applying the habitable zone filter alone, without introducing feature weighting.

Experiment #1: Baseline Ranking Comparison Study						
	Model #1	Model #2	Model #3	Model #4	Model #5	Model #6
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted Baseline Manhattan	Unweighted Baseline Euclidean	Unweighted Habitable Zone Only Baseline Manhattan	Unweighted Habitable Zone Only Baseline Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-438 b	Kepler-438 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-296 e	Kepler-442 b
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-132 e	Kepler-442 b	Kepler-395 c
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-395 c	Kepler-296 e
5	Kepler-1701 b	Kepler-1701 b	Kepler-384 c	Kepler-130 d	Kepler-138 e	Kepler-138 e
6	Kepler-22 b	Kepler-1126 c	Kepler-1638 b	Kepler-1440 b	Kepler-452 b	Kepler-452 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1498 b	Kepler-1410 b	Kepler-1410 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1965 b	Kepler-1441 b	Kepler-1544 b	Kepler-1544 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1440 b	Kepler-1965 b	Kepler-1701 b	Kepler-1701 b
10	Kepler-69 c	Kepler-311 d	Kepler-1605 b	Kepler-1620 b	Kepler-1606 b	Kepler-1450 b
Top 5 overlap between weighted proposed method & unweighted baseline			1/5	0/5	0/5	0/5
Top 10 overlap between weighted proposed method & unweighted baseline			3/10	1/10	3/10	3/10

Table 1: Displays Experiment #1 Baseline Study Models which compares the Weighted Proposed Method KNN Models #1 and #2 with Unweighted Baseline KNN Models #3, #4, #5, and #6

The results of the Experiment #1: Baseline Ranking Comparison Study referenced in Table 1. When comparing the Weighted Proposed Method Manhattan KNN Model #1 with Unweighted Baseline Manhattan KNN Model #3 to determine the effects of feature weighting, only a small portion of top Earth-Like exoplanets remained the same when weights were removed. Only Kepler-452 b remained the most Earth-Like when weights were removed however no one top 5 exoplanet remained consistent. Overall, 3 exoplanets remained in the top 10 list. These results indicate that the features weights proposed are not redundant but rather strongly influence final rankings. These results also indicate that unweighted Manhattan distance may overvalue some parameters or define all parameters as equally relevant for Earth-Like similarity and potential habitability. Our proposed method, however, better prioritizes Earth-relevant parameters.

A similar trend can be observed when comparing the Weighted Proposed Method Euclidean KNN Model #2 with Unweighted Baseline Euclidean KNN Model #4. When features weights were removed, none of the top 5 exoplanets remained and only 1 exoplanet remained in the top 10 list, showing more significant changes compared to the Manhattan distance metric results. This shows that the Euclidean distance metric is highly sensitive to scaling and feature weighting. Therefore, the proposed weighted Euclidean distance may be able to correct the distortions caused by the unweighted Euclidean distance, resulting in a more stabilized ranking.

When comparing the Weighted Proposed Method Manhattan KNN Model #1 with the Unweighted Habitable Zone Only Baseline Manhattan KNN Model #5, none of the top 5 exoplanets remained consistent however only 3 exoplanets remained the same in both top 10 lists. The same results appear when comparing the Weighted Proposed Method Euclidean KNN Model #2 with the Unweighted Habitable Zone Only Baseline Euclidean KNN Model #6. These observations show that simple habitable zone filtering does not produce the same results as our proposed weighted rankings. These results indicate that only filtering for exoplanets who are within the habitable zone of their host stars alone is not sufficient in determining Earth-Like similarity. This shows that other relevant Earth-Like parameters such as planetary radius and temperature, orbital distance, orbital period, and stellar properties are equally important. Therefore, our proposed weighted features capture multi-factor Earth similarity rather than just location in orbit.

An important observation to note is that the exoplanets Kepler-452 b appears as the most robust candidate since it remained the most consistent across all ranking methods except for Unweighted Baseline Euclidean KNN Model #4, where its ranking dropped to 15th place. These results verify that Kepler-452 b is the closest match to Earth with a potential for habitability. In addition to Kepler-452 b, the exoplanets Kepler-1638 b, Kepler-1126 c, Kepler-1606 b, Kepler-1701 b, Kepler-1185 b, and Kepler-442 b appeared most frequently across various ranking methods showing that these exoplanets are relatively robust candidates.

Experiment #2: Ablation Study

The purpose of the Experiment #2: Ablation Study is to identify which feature groups are most important for the Earth-like ranking. The same unweighted ranking method will be run for both the Manhattan and Euclidean distance metrics but each time, a different group of features will be removed to observe how the top Earth-Like exoplanets change and to determine which features are the most relevant for Earth-Like similarity ranking.

The Weighted Proposed Method Manhattan KNN Model #7 and Weighted Proposed Method Euclidean KNN Model #8 represent the Full Model group where all features are selected with proposed weights applied. All other models for the Experiment #2: Ablation Study will be compared to KNN Models #7 and #8 based on the Manhattan and Euclidean distance metric respectively.

The No Spectral Type group is represented by the Unweighted No Spectral Type Manhattan KNN Model #9 and Unweighted No Spectral Type Euclidean KNN Model #10 where only the one hot encoded spectral type columns are removed. The No Stellar Features (Partial) group is represented by the Unweighted No Stellar Features (Partial) Manhattan KNN Model #11 and Unweighted No Stellar Features (Partial) Euclidean KNN Model #12 where all stellar features are removed but one hot encoded spectral type features are still included. The stellar features removed include stellar mass, stellar radius, stellar effective temperature, stellar luminosity, and stellar surface gravity. Likewise, the No Stellar Features (Full) group is represented by the Unweighted No Stellar Features (Full) Manhattan KNN Model #11.5 and Unweighted No Stellar Features (Full) Euclidean KNN Model #12.5 where all stellar and one hot encoded spectral type features are removed.

The No Planetary Features group is represented by the Unweighted No Planetary Features Manhattan KNN Model #13 and Unweighted No Planetary Features Euclidean KNN Model #14. The planetary features removed include planet radius, planet mass, planet equilibrium temperature, and planetary orbital velocity. The No Orbital Features group is represented by the Unweighted No Orbital Features Manhattan KNN Model #15 and Unweighted No Orbital Features Euclidean KNN Model #16. The orbital features removed include orbital period and orbital distance (orbital semi major axis). The No Habitable Zone group is represented by the Unweighted No Habitable Zone Manhattan KNN Model #17 and Unweighted No Habitable Zone Euclidean KNN Model #17 where the habitable zone binary column is removed. Lastly, the No Temperature group is represented by the Unweighted No Temperature Manhattan KNN Model #19 and Unweighted No Temperature Euclidean KNN Model #20 where the planetary equilibrium temperature and stellar effective temperature features were removed.

Experiment #2: Ablation Study								
	Full Model		No Spectral Type		No Stellar Features (Partial)		No Stellar Features (Full)	
	Model #7	Model #8	Model #9	Model #10	Model #11	Model #12	Model #11.5	Model #12.5
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted No Spectral Type Manhattan	Unweighted No Spectral Type Euclidean	Unweighted No Stellar Features Manhattan (Partial)	Unweighted No Stellar Features Euclidean (Partial)	Unweighted No Stellar Features Manhattan (Full)	Unweighted No Stellar Features Euclidean (Full)
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1185 b	Kepler-442 b	Kepler-442 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-132 e	Kepler-438 b	Kepler-186 f
3	Kepler-51 e	Kepler-51 e	Kepler-442 b	Kepler-132 e	Kepler-1605 b	Kepler-442 b	Kepler-186 f	Kepler-438 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-132 e	Kepler-1620 b	Kepler-442 b	Kepler-186 f	Kepler-452 b	Kepler-138 e
5	Kepler-1701 b	Kepler-1701 b	Kepler-1126 c	Kepler-384 c	Kepler-438 b	Kepler-1253 b	Kepler-138 e	Kepler-1185 b
6	Kepler-22 b	Kepler-1126 c	Kepler-438 b	Kepler-130 d	Kepler-132 e	Kepler-1498 b	Kepler-395 c	Kepler-395 c
7	Kepler-1126 c	Kepler-69 c	Kepler-1620 b	Kepler-1440 b	Kepler-1126 c	Kepler-438 b	Teegarden's Star d	Kepler-1229 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-384 c	Kepler-1498 b	Kepler-384 c	Kepler-384 c	Teegarden's Star c	Teegarden's Star d
9	Kepler-1690 b	Kepler-442 b	Kepler-1638 b	Kepler-442 b	Kepler-1965 b	Kepler-1965 b	TOI-700 e	TOI-700 d
10	Kepler-69 c	Kepler-311 d	Kepler-1253 b	Kepler-1459 b	Kepler-186 f	Kepler-138 e	TOI-700 d	GJ 1002 c
Top 5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	1/5	0/5	0/5	0/5
Top 10 overlap between weighted proposed method & unweighted ablation			3/10	2/10	2/10	2/10	1/10	2/10

Table 2: Displays Experiment #2 Ablation Study Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Unweighted Ablation KNN Models #9, #10, #11, #12, #11.5, and #12.5

The results of the Experiment #1: Ablation Study referenced in Table 2 and Table 3. The results in Table 2 show that when comparing the Full Model to the No Spectral Type group, there appears to be significant changes in both the Manhattan and Euclidean distances when the spectral type columns are removed. This implies that the spectral type features contribute significantly to Earth-Like ranking stability. When comparing the Full Model to the No Stellar Feature (Partial) group, the results show that even with spectral type features remaining, removing stellar mass, radius, temperature, luminosity, and gravity cause major ranking disruption. This suggests continuous stellar variables are important features for determining Earth-Like similarity. When comparing the Full Model to the No Stellar Feature (Full) group, the results show that removing all stellar and spectral type information causes the strongest disruptions in similarity rankings. This implies that the host star properties are among the most critical determinants of Earth-like similarity. This can be expected since the stellar features paired with the spectral type, describing star type and luminosity class, provide strong insight into the stellar lifespan, solar flare activity, radiation environment, long-term planetary climate stability, and habitability zone location. These are all critical parameters for Earth-Like surface composition and potential habitability.

Experiment #2: Ablation Study										
	Full Model		No Planetary Features		No Orbital Features		No Habitable Zone		No Temperature	
	Model #7	Model #8	Model #13	Model #14	Model #15	Model #16	Model #17	Model #18	Model #19	Model #20
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted No Planetary Features Manhattan	Unweighted No Planetary Features Euclidean	Unweighted No Orbital Features Manhattan	Unweighted No Orbital Features Euclidean	Unweighted No Habitable Zone Manhattan	Unweighted No Habitable Zone Euclidean	Unweighted No Temperature Manhattan	Unweighted No Temperature Euclidean
1	Kepler-452 b	Kepler-452 b	HD 43197 b	Kepler-1625 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-1185 b	Kepler-452 b	Kepler-1570 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1638 b	Kepler-1638 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1253 b	Kepler-1185 b	Kepler-1253 b
3	Kepler-51 e	Kepler-51 e	Kepler-452 b	Kepler-784 b	Kepler-384 c	Kepler-384 c	Kepler-132 e	Kepler-132 e	Kepler-1605 b	Kepler-1185 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-289 e	Kepler-1570 b	Kepler-132 e	Kepler-132 e	Kepler-1126 c	Kepler-384 c	Kepler-384 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-1625 b	Kepler-1929 b	Kepler-1965 b	Kepler-1440 b	Kepler-384 c	Kepler-130 d	Kepler-1440 b	Kepler-1440 b
6	Kepler-22 b	Kepler-1126 c	PH2 b	Kepler-525 b	Kepler-1440 b	Kepler-1498 b	Kepler-1253 b	Kepler-1440 b	Kepler-1570 b	Kepler-132 e
7	Kepler-1126 c	Kepler-69 c	Kepler-1634 b	Kepler-638 b	Kepler-1253 b	Kepler-130 d	Kepler-1965 b	Kepler-1498 b	Kepler-132 e	Kepler-1605 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1636 b	HD 43197 b	Kepler-1505 b	Kepler-1441 b	Kepler-1440 b	Kepler-1441 b	Kepler-1253 b	Kepler-130 d
9	Kepler-1690 b	Kepler-442 b	HD 142415 b	HD 44219 b	Kepler-1498 b	Kepler-1965 b	Kepler-1605 b	Kepler-1965 b	Kepler-1505 b	Kepler-1441 b
10	Kepler-69 c	Kepler-311 d	Kepler-1519 b	Kepler-806 b	Kepler-1441 b	Kepler-1505 b	Kepler-1505 b	Kepler-1620 b	Kepler-1965 b	Kepler-1498 b
Top 5 overlap between weighted proposed method & unweighted ablation			2/5	1/5	1/5	0/5	1/5	0/5	1/5	0/5
Top 10 overlap between weighted proposed method & unweighted ablation			2/10	1/10	1/10	1/10	2/10	1/10	1/10	1/10

Table 3: Displays Experiment #2 Ablation Study Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Unweighted Ablation KNN Models #13, #14, #15, #16, #17, #18, #19, and #20.

The results in Table 3 show that when comparing the Full Model to the No Planetary Features group, there appears to be strong rankings shifts when planetary features are removed. However, its effects are surprisingly less significant than when all stellar and spectral type features are removed. This implies that both planet and host star features are jointly critical for determining Earth-Like similarity rather than planetary features alone.

Comparing the Full Model to the No Orbital Features group and the No Temperature group respectfully reveals that removing orbital or temperature features had equally the strongest effects on ranking stability due to the low overlap values for the top 5 and top 10 lists. This can be expected since orbital properties along with planetary and stellar temperatures are critical for determining habitability conditions and planet atmospheric composition. Additionally, given only about 1.2% of all confirmed exoplanets are found to have an orbital period similar to Earth's between 300-400 days, orbital period is therefore one of the most rare and distinctive features needed for Earth-Like similarity assessment. This effect may also be influenced by detection bias, as

longer period exoplanets remain underrepresented in current catalogs. Lastly, comparing the Full Model to the No Habitable Zone group revealed that removing this binary feature causes notable changes to the top 5 and top 10 results. This shows that habitable zone contributes meaningful ranking power but is not sufficient alone based on Experiment #1 results.

Both Table 2 and Table 3 show that the top 5 overlap and top 10 overlap remain very low (between 0-3 overlapping planets) across all ranking methods when compared to full weighted model. This indicates that Earth-Like similarity ranking depends on multiple interacting feature groups rather than a single feature or feature group. This therefore supports the validity of the proposed method for a weighted full model.

The features which had the highest importance were orbital features, temperature features, and host star features while planetary features and habitable zone had strong importance. This indicates that host star environment and thermal conditions are central to Earth similarity assessment. Kepler-452 b once again remained the most robust in rankings followed by Kepler-1638 b, Kepler-1126 c, Kepler-1185 b, and Kepler-442 b which supports the results from Experiment #1.

Across nearly all ablations, Manhattan distance retained more overlap and thus, exhibited greater ranking stability than Euclidean distance. This can be expected due to the sensitivity of the Euclidean distance metric. This suggests improved robustness for combined exoplanet feature similarity.